

Apple Inc.

Supply Chain Diversification Strategy Review 2025

Prepared for: Apple Inc. Board of Directors

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01 | Executive Summary

RECOMMENDATION: Accelerate supply chain diversification via a phased 3-year shift — targeting 20% of iPhone assembly in India by 2027 — while investing in supplier ecosystem development to reduce unit cost premium from ~\$12 to ~\$6 per device.

4.2/5.0

Risk Score

China concentration risk

\$5.8B

Cost Delta

Annual cost at 20% shift

-17%

Risk Reduction

Tariff exposure at 20% shift

Key Findings

~88% of iPhone assembly concentrated in China — highest geopolitical risk of any Apple product line

- Trade policy escalation could cost Apple \$8-12B annually if unmitigated; diversification payback within 4 years
- India is the only viable large-scale alternative; Vietnam capacity limited to non-iPhone products near-term
- Taiwan semiconductor dependency (TSMC) is a separate, higher-urgency risk requiring parallel action

Immediate Next Steps

1. Commit to India Phase 1 investment (\$2.1B capex, FY2025)
2. Establish supplier development fund for Indian tier-2/3 ecosystem
3. Accelerate TSMC Arizona ramp — seek anchor customer pricing
4. Create board-level Supply Chain Risk Committee

02 | Problem Framing

Apple's supply chain is structurally exposed to three compounding risks

HIGH

Trade Policy Risk

US tariffs on Chinese electronics now at levels that compress iPhone margins by 2–4pp. Escalation scenarios modeled at up to \$12B annual impact. No near-term resolution visible.

CRITICAL

Taiwan / Semiconductor Risk

TSMC produces 100% of Apple's custom silicon. Cross-strait tensions at a decade high. Arizona fabs not at scale until 2027+. A Taiwan crisis = no iPhone production.

HIGH

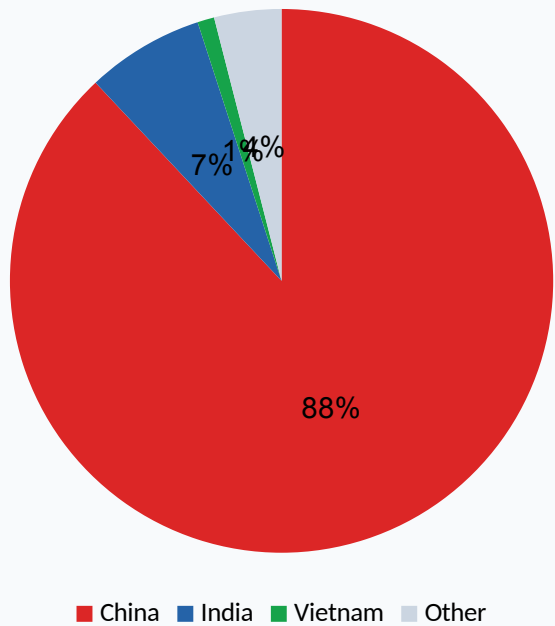
Operational Concentration Risk

iPhone City (Zhengzhou) assembles >50% of global iPhone output. One facility. COVID lockdowns in 2022 cost Apple ~\$8B in delayed holiday revenue.

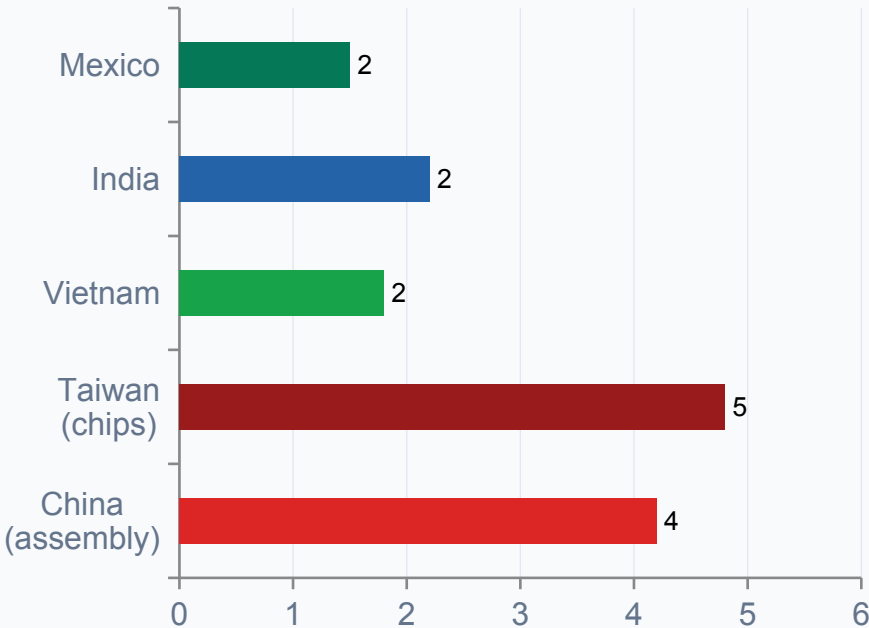
Bottom line: These risks are not independent — a Taiwan crisis triggers both semiconductor AND assembly disruption simultaneously.

03 | Data Analysis — Manufacturing Concentration

iPhone Assembly by Geography (2025)



Supplier Risk Score by Geography (1–5)



Source: US Import Trade Records (ImportYeti), Apple Supplier Responsibility Report 2025, analyst estimates. Risk scores are composite weighted averages across 5 dimensions.

03 | Data Analysis — Supplier Risk Scorecard

Weighted risk scoring across five strategic dimensions — China scores 4.2/5.0, highest of any geography

Risk Dimension	Weight	China	India	Vietnam
Geopolitical exposure	40%	5 	2	2
Supply concentration	25%	5 	2	3
Lead time / logistics	15%	2	3	2
Labor & ops stability	10%	3	2	2
Ecosystem maturity	10%	1	3	3
Composite score	100%	4.2	2.4	2.3

 = scores ≥ 4 flagged as high risk. Composite = weighted average. China ecosystem maturity (score 1) reflects deep tier-2/3 supplier networks built over 20 years.

04 | Options Evaluated

Three geographies assessed for iPhone manufacturing diversification

RECOMMENDED

India

Strengths

Largest viable labor pool (1.4B)

-
- Government PLI subsidies
-
- English-speaking engineering talent
-
- Strategic political relationship

Constraints

Supplier ecosystem 8–10 yrs behind China

-
- Infrastructure constraints (power, ports)
-
- +\$9–14/unit cost premium initially

PARTIAL FIT

Vietnam

Strengths

Low labor costs

-
- Established Apple presence (AirPods, Watch)
-
- Proximity to Chinese suppliers
-
- Stable government

Constraints

Cannot absorb iPhone complexity at scale

-
- Smaller labor pool limits max capacity
-
- High dependency on Chinese components

LONGER TERM

Mexico

Strengths

USMCA tariff advantages

-
- US market proximity (2-day shipping)
-
- Growing electronics ecosystem
-
- Nearshoring trend momentum

Constraints

Early-stage ecosystem for Apple-grade mfg

-
- Significant labor cost premium vs. Asia
-
- Limited high-precision assembly experience

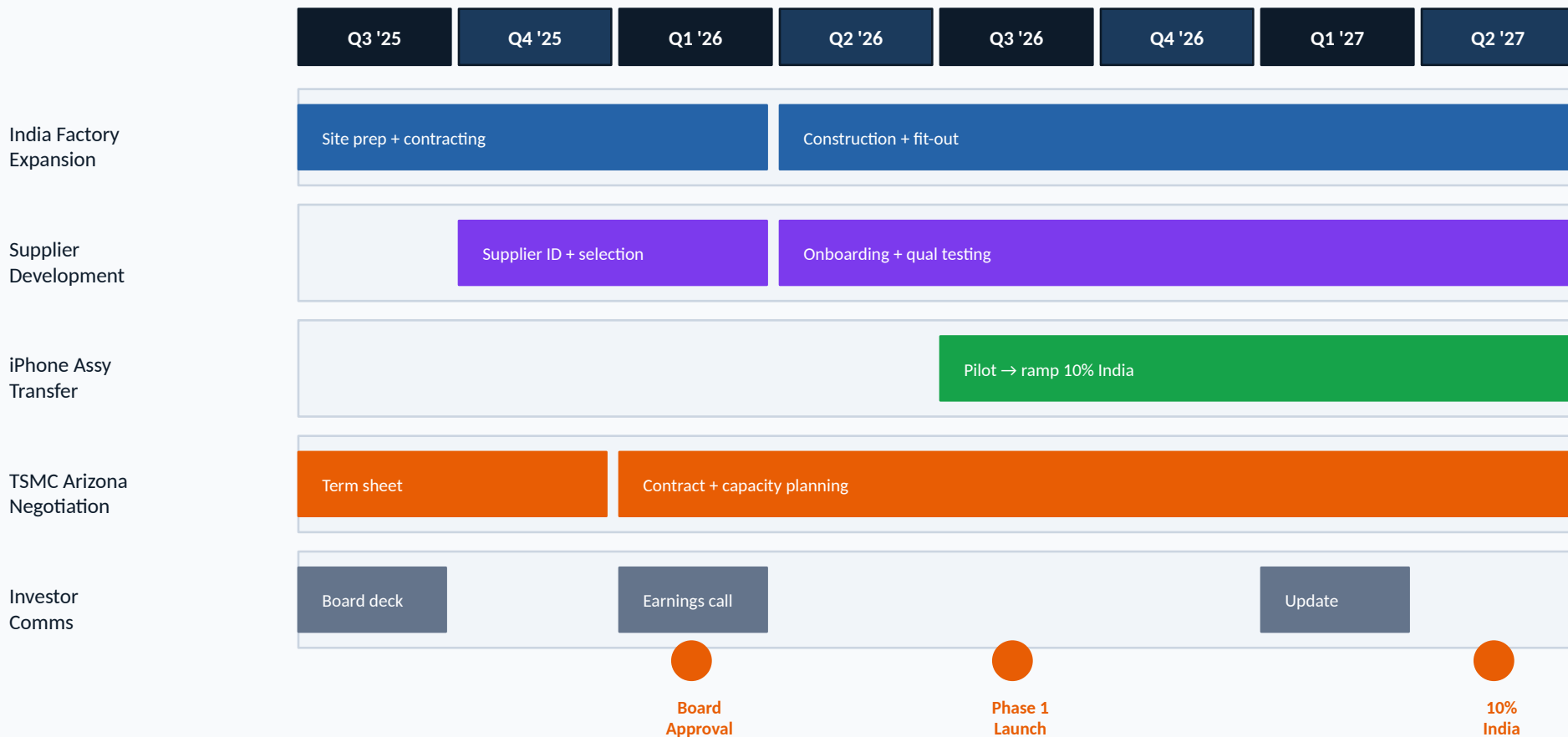
05 | Recommended Path

A phased diversification — not a sudden shift — that builds ecosystem capability while reducing concentration risk

Phase 1 Foundation	Phase 2 Scale	Phase 3 Resilience
2025–2026	2026–2027	2027–2028
Commit \$2.1B capex to India expansion (Foxconn + Tata facilities) • • Establish Apple Supplier Development Fund (\$500M) to build tier-2/3 Indian ecosystem • • Target: 10% iPhone assembly in India by end of 2026 • • Begin TSMC Arizona anchor customer negotiations	Ramp India iPhone assembly to 20% of global volume • • Onboard 15–20 critical Indian tier-2 component suppliers • • Expand Vietnam non-iPhone capacity by 40% • • TSMC Arizona Phase 1 fab online — initial A-series chip production	Assess Mexico feasibility for Mac Pro / desktop lines • • Target China share below 70% for iPhone assembly • • Full TSMC Arizona ramp — 15–20% of Apple chip volume • • Annual supply chain risk review formalized at board level

Target outcome by 2028: China share of iPhone assembly <70% | Composite risk score reduced from 4.2 → 3.1 | Tariff exposure reduced by 25%

06 | Implementation Roadmap



07 | Financial Model — Cost of Diversification

Diversification carries real near-term costs — but tariff savings and risk reduction justify the investment by Year 4

\$2.1B

Phase 1 Capex

India factory investment

\$5.8B

Peak Annual Cost

At 20% shift (Year 3)

~Yr 4

Payback Period

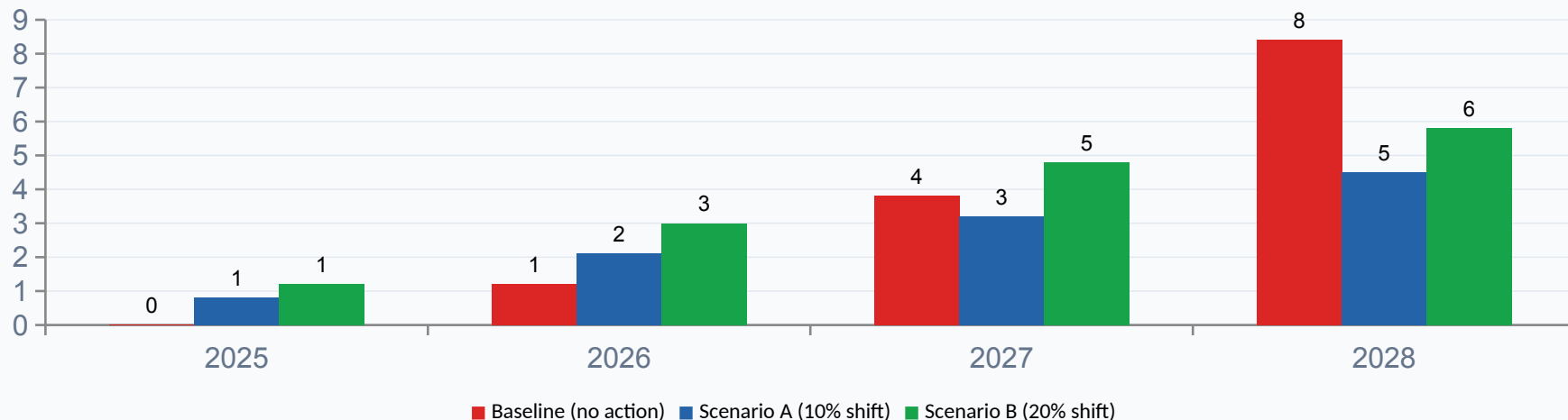
vs. tariff escalation scenario

25%

Risk Reduction

Composite score improvement

Cumulative Cost Impact vs. Tariff Escalation (\$B) — Scenarios 2025–2028



08 | Risks & Mitigations

Risk	Likelihood	Impact	Mitigation
India quality ramp slower than planned	Medium	High	Maintain China backup capacity; stage ramp tied to quality KPIs
Tariff truce makes diversification look expensive	Low	Medium	Frame as resilience investment; risk reduction has standalone value
TSMC Arizona delayed beyond 2027	Medium	Critical	Accelerate TSMC Japan capacity; explore Samsung foundry as backup
Foxconn India labor/operational issues	Medium	High	Dual-source with Tata Electronics; contractual penalties for shortfall
China retaliatory trade measures vs. Apple	Low	High	Legal / diplomatic monitoring; scenario plan with comms team
Cost overruns on supplier dev fund	Medium	Medium	Milestone-gated disbursements; quarterly operations review

The window to act is now.

Every quarter Apple delays diversification, the cost of the eventual transition increases — and the probability of a forced, crisis-driven move grows.

Approve Phase 1 India investment (\$2.1B) at the September board meeting

-
- Establish Supplier Development Fund and governance structure by Q4 2025
-
- Authorize TSMC Arizona anchor capacity negotiations immediately
-
- Communicate supply chain resilience strategy to investors at December earnings call